

Use Ratio Reasoning

Lesson 3-6

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.


$$\frac{1}{2} = \frac{2}{4}$$

$$\frac{2}{3} = \frac{8}{12}$$

Proportion

Write the definition:


$$\frac{1}{2} = \frac{2}{4}$$

$$\frac{2}{3} = \frac{8}{12} \rightarrow 2 \times 12 = 24 \text{ and } 3 \times 8 = 24 \checkmark$$

Cross-multiply

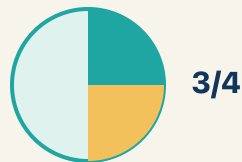
Write the definition:


$$\times 2$$

$$5:8 \times 3 \rightarrow 15:24$$

Scale

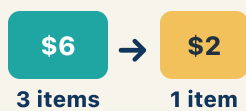
Write the definition:



$\frac{1}{2}$ and $\frac{3}{6}$ and $\frac{5}{10}$ all equal the same amount — half

Equivalent

Write the definition:



60 miles in 3 hours $\rightarrow$ 20 miles per 1 hour

Unit rate

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can use ratio reasoning to solve problems with proportions and scaling.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Proportion

Cross-multiply

Scale

Equivalent

Unit rate

1 An equation that states two ratios are equal is a _____.

2 Multiplying the numerator of one ratio by the denominator of the other to check a proportion is to _____.

3 To multiply a ratio up or down by the same factor is to _____ it.

4 Two ratios that show the same comparison are _____.

5 The amount for exactly one unit is the _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A recipe uses 4 cups of broth for every 10 servings. How many cups of broth are needed for 30 servings?

- A. 12
- B. 8
- C. 14
- D. 40

Step 1 Scale factor: $30 \div 10 = 3$.

Step 2 Multiply broth by 3: $4 \times 3 = 12$ cups.

 **Answer:** A. 12

 We do — together

Solve this one with your class using the same steps.

Problem: Are the ratios 6:9 and 2:3 equivalent?

- A. Yes, both simplify to 2:3
- B. No, $6 \neq 2$
- C. Yes, because $6 + 9 = 15$
- D. No, because $9 - 6 \neq 3 - 2$

Step 1 _____

Step 2 _____

Answer: _____

**You do — your turn**

Now try one on your own. Show every step.

Problem: If 3 pounds of apples cost \$6, how much do 7 pounds cost?

A. \$14

B. \$13

C. \$10

D. \$21

Show your work:

| Try It

Solve on your own. Check the answer key when you are done.

1. A marinara recipe uses 4 cups of tomatoes for every 6 servings. The chef needs 18 servings. How many cups of tomatoes should she use?

- A. 12 cups
- B. 10 cups
- C. 16 cups
- D. 9 cups

Show your work:

2. During a cooking-team relay, a chef plates 3 finished dishes for every 5 minutes she works. At the same rate, how many dishes will she plate in 20 minutes?

- A. 12 dishes
- B. 10 dishes
- C. 15 dishes
- D. 8 dishes

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A recipe serves 6 people and uses 4 cups of rice. You need to serve 15 people. Show two different methods to find how much rice you need: (1) using a scale factor and (2) using a unit rate.

Show your work:

Reflect — Exit Ticket

A recipe calls for 6 cups of flour for every 15 cookies. How many cups of flour are needed to make 45 cookies?

- A. 18
- B. 12
- C. 24
- D. 21

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Proportion
2. **Notes 2:** Cross-multiply
3. **Notes 3:** Scale
4. **Notes 4:** Equivalent
5. **Notes 5:** Unit rate
6. **We Try (notes):** A. Yes, both simplify to 2:3 — *Divide 6:9 by 3 to get 2:3. Since both ratios simplify to 2:3, they are equivalent.*
7. **You Try (notes):** A. \$14 — *Unit rate: $\$6 \div 3 = \2 per pound. For 7 pounds: $\$2 \times 7 = \14 .*
8. **Try It 1:** A. 12 cups — *Scale factor: 18 servings divided by 6 servings = 3. Multiply the tomatoes by 3: $4 \times 3 = 12$ cups.*
9. **Try It 2:** A. 12 dishes — *Scale factor: 20 minutes divided by 5 minutes = 4. Multiply the dishes by 4: $3 \times 4 = 12$ dishes.*
10. **Exit Ticket:** A. 18 — *Scale factor: $45 \div 15 = 3$. Multiply flour by 3: $6 \times 3 = 18$ cups of flour.*